WHITEPAPER

Functional Equivalence and the Limits of Law:

How Artificial Intelligence Will Fundamentally Disrupt Copyright and Patent Law

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Abstract

Current intellectual property law operates on assumptions rooted in human cognitive limitations: that copying requires significant effort, that translation between forms requires skill and time, and that substantial similarity can be assessed by comparing surface-level expression. Artificial intelligence shatters all three assumptions simultaneously. This paper examines a specific and timely scenario — the unauthorized disclosure of proprietary source code followed by rapid AI-assisted reimplementation in functionally identical but syntactically distinct languages — to argue that copyright and patent law are structurally unprepared for the AI era. We analyze the doctrinal gaps in existing law, explore the ethical dimension that law has historically ignored, and propose a framework for reform centered on the concept of "functional equivalence" as a legally cognizable harm. The central thesis is this: when AI can convert any expression into any other expression while preserving all functional meaning, the classical distinction between idea and expression collapses, and intellectual property law must be fundamentally reconceived — not merely amended.

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Executive Summary

The Scenario

On April 1, 2026, proprietary source code from a major AI platform became publicly accessible through an inadvertent exposure of TypeScript source map files — debugging artifacts left reachable on production infrastructure. Within hours the code was forked. Within 48 hours, AI systems had rewritten it in Python. Within days, a second AI pass produced a Rust implementation. The Rust code was compiled to a native binary, debug symbols stripped, and the resulting executable made available — a fully functional equivalent of a multi-year, hundred-million-dollar software investment, with no line of the original code visible anywhere in the artifact that shipped.

Most people's instinct is that this is obviously illegal. That instinct is ethically correct. Legally, it is far more complicated — and the complication reveals a structural failure in intellectual property law that artificial intelligence has converted from a theoretical vulnerability into an operational weapon.

The Legal Reality

Copyright law protects expression, not function. When AI rewrites code in a different language, it produces entirely new expression while preserving all function — defeating the primary test for infringement. Patent law protects function directly, but software patents require two to four years to obtain, cost $15,000 to $50,000 to prosecute, and are routinely invalidated under the Supreme Court's 2014 Alice decision. By the time a patent issues, the harm is complete. Trade secret law provides the strongest existing claim — but only against actors who knew the information was improperly disclosed, only for as long as the secret remains meaningful, and only against defendants amenable to US jurisdiction.

The deeper problem is structural. AI has destroyed the three assumptions on which all IP doctrine rests: that copying requires effort, that translation between forms requires skill and time, and that similarity can be assessed by comparing surface expression. When a compiled binary is all that ships, there is no surface expression to compare. When AI performs the translation in hours, there is no effort to deter. When the reimplementation chain runs TypeScript to Python to Rust to binary across four jurisdictions, there may be no infringer amenable to any single court's authority. The doctrinal framework does not bend under this pressure. It collapses.

The Strategic Reality

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This is not merely a legal problem. It is a national security problem that IP law has been asked to solve — and AI has made the inadequacy of that substitution impossible to ignore.

North Korea's Lazarus Group — a state operation controlled by Pyongyang's Reconnaissance General Bureau — stole $2.02 billion in cryptocurrency in 2025 alone using pre-AI cyber capabilities, representing approximately 60% of all global cryptocurrency theft that year. A UN Security Council report found these operations generate roughly half of North Korea's foreign currency income and fund an estimated 40% of its weapons of mass destruction programs. The February 2025 Bybit exchange compromise — $1.5 billion in Ethereum, the largest single financial theft in recorded history — was executed through exploitation of a software supply chain vulnerability, laundered at a speed blockchain analysts described as previously unimaginable. This is what state-sponsored economic warfare looks like before AI amplification.

AI multiplies that capability by orders of magnitude while collapsing its cost, extending it from the handful of nations with North Korea's extraordinary cyber investment to any state actor with political will and computing infrastructure. A nation with 25 million people and a $20 billion GDP — unable to field a competitive military, unable to project conventional force beyond its borders — can now operate an AI-enabled economic warfare capability that inflicts damage on a $25 trillion economy measurable in hundreds of billions annually. The military-economic correlation that stabilized the post-WWII international order for seventy years has been severed. The international order has not yet noticed.

In 1999, two senior colonels of the People's Liberation Army published "Unrestricted Warfare" — a strategic doctrine arguing that a technologically inferior nation could defeat a militarily superior opponent by conducting warfare across every non-kinetic domain simultaneously while remaining permanently below the threshold that triggers recognized military response. That doctrine has found its weapon. AI-enabled economic warfare is unrestricted warfare with precision instruments — invisible, deniable, faster than any legal or diplomatic response, and exploiting a gap in the international legal framework that was written before the capability existed.

What This Paper Argues

This paper traces the logical chain from a specific IP law gap to its terminus in geopolitical security. It argues three things simultaneously:

- Intellectual property law must be fundamentally reformed — not amended at the margins — to address AI-enabled functional reimplementation. The specific proposals are a Fast-Track Functional Architecture Registration system, expedited injunctive relief

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procedures, mandatory AI provenance tracking, and a functional copyright doctrine protecting software architecture against language-agnostic reproduction. These reforms address the commercial misappropriation threat.

- For state-sponsored actors, IP law was never the right framework and reform alone is insufficient. The appropriate response toolkit operates through export controls under IEEPA and the Export Administration Regulations, CFIUS review, the Economic Espionage Act, and — most urgently — explicit threshold recognition of AI-enabled economic warfare as a category of aggression under international law, parallel to what the Tallinn Manual attempted for kinetic cyber operations.
- The international community faces a pre-framework moment analogous to 1945 — when the dominant technology of conflict outran the institutional capacity to govern it — but without the catastrophic visibility that forced action after Hiroshima. AI economic warfare leaves no mushroom cloud. Its damage accumulates quietly, company by company, sector by sector, below every threshold of recognition and response, until the barn door question is moot and the horses are gone.

Structure of This Paper

This paper proceeds in thirteen sections. Sections 1 through 3 analyze the existing legal landscape across copyright, patent, and trade secret law. Section 4 addresses the patent cost barrier and the structural exclusion of small innovators from the system's protective function. Sections 5 through 7 examine the ethical dimension of the law-ethics gap, the AI-specific legal challenges of attribution and velocity, and the proposed functional equivalence framework. Section 8 analyzes the political economy of IP reform. Section 9 addresses the technical dimensions — source map vulnerability, AI language translation capability, and the compiled binary enforcement void. Section 10 examines the compiled binary problem as the paper's most technically original contribution: the recognition that all proposed enforcement mechanisms require source code visibility that compiled distribution deliberately eliminates. Section 11 addresses the governance gap and the non-law toolkit — platform governance, AI provider policy, and provenance standards — that must operate while legal reform develops. Section 12 develops the geopolitical argument in full: AI as the equalizer of state power projection, the North Korea model generalized, the Unrestricted Warfare doctrine weaponized, and the three unanswered questions international law must resolve. Section 13 presents conclusions stratified by timeframe and the paper's ultimate finding.

Ultimate Finding

AI has made economic warfare as potentially existential as kinetic warfare — while making it simultaneously cheaper, faster, more deniable, and less governable than any form of conflict in human history. Intellectual property law is not equipped to address this.

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The international legal order is not equipped to recognize it. The governance frameworks that protect innovation, sustain competitive markets, and underwrite geopolitical stability were all built for a world in which this capability did not exist. That world is gone.

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1. Introduction: The Scenario That Breaks Existing Law

On April 1, 2026, a significant quantity of proprietary source code — allegedly originating from Anthropic's Claude Code platform — became publicly accessible through an inadvertent exposure of TypeScript source map files. These map files, intended as debugging aids, contained human-readable representations of minified or compiled JavaScript/TypeScript code. Within hours, the code was forked on public repositories. Within days, AI systems had rewritten it — first in Python, then in Rust — producing functionally equivalent implementations in languages wholly distinct from the original.

This scenario presents intellectual property law with a challenge it was not designed to meet. A sophisticated human programmer could, in theory, accomplish the same outcome — but only over weeks or months, involving deep reading, comprehension, architectural analysis, and deliberate reimplementation. That temporal friction historically served as a practical deterrent, and courts have long assumed it as a background condition when crafting doctrine. AI eliminates that friction entirely. What once required months now requires hours. What once required a team now requires a prompt.

> **The Core Problem**
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> The law has never had to confront a technology that can instantaneously translate any expression into a functionally identical but formally distinct expression. Copyright protects expression, not function. Patent law protects function, but only for hardware and process inventions — not software logic in most jurisdictions. AI-assisted reimplementation falls into the gap between both.

The immediate instinct — shared by many non-lawyers — is to say that this reimplementation is clearly illegal. After all, the originating code was proprietary, accessed without authorization, and the resulting work reproduces its entire functional architecture. This instinct is ethically sound. Legally, however, the situation is far murkier, and the murkiness reveals deep structural problems in our IP framework that AI has merely made impossible to ignore.

2. The Current Legal Landscape: Copyright Law

2.1 The Idea-Expression Dichotomy

Copyright law in the United States, and in most jurisdictions following the Berne Convention, protects the expression of ideas but not ideas themselves. This principle — codified in 17 U.S.C. § 102(b) in the U.S. — is the bedrock of copyright doctrine. A novelist cannot copyright the concept of a dystopian surveillance state; she can only copyright her specific words. A

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software developer cannot copyright the algorithm for sorting a list; she can only copyright the specific code she wrote to implement it.

This distinction has always been philosophically unstable at the margins, and courts have struggled with it in software cases for decades. The Ninth Circuit's analysis in Oracle America, Inc. v. Google LLC (before the Supreme Court's resolution on fair use grounds) grappled with whether the structure, sequence, and organization of an API constituted protectable expression or unprotectable functionality. The Second Circuit's seminal Computer Associates v. Altai (1992) decision introduced the "abstraction-filtration-comparison" test precisely because software blurs the line between idea and expression at every level of analysis.

> **Altai's Abstraction-Filtration-Comparison Test**
>
> Under Altai, courts analyze software at multiple levels of abstraction (from general purpose down to individual code lines), filter out elements that are unprotectable (ideas, elements dictated by efficiency, public domain material, and externally imposed specifications), and then compare whatever expression remains. AI-generated reimplementation defeats this test by ensuring that nothing in the filtration stage's "expression" layer survives to the comparison stage — because every line has been rewritten.

When AI rewrites code in a different language, what is left to compare? The variable names are different. The syntax is different. The idiomatic patterns are different. The file structure may be entirely reorganized. If a court applies the Altai test faithfully, it may find no protectable expression remaining after filtration — because everything that survived filtration as "pure expression" has been transformed. The function is identical. The code is not.

2.2 The Clean Room Defense and Its AI-Era Collapse

Courts have long recognized a "clean room" defense in software copyright cases. If a defendant can demonstrate that its implementation was developed by engineers who had no access to the plaintiff's code — that they worked only from a functional specification — copyright infringement is typically not established. The paradigmatic example is the IBM PC clone manufacturers of the 1980s, who used clean room techniques to reproduce the BIOS firmware without copying IBM's expression.

The clean room defense rests on a practical assumption: that deriving a functional specification from an implementation, and then re-implementing from that specification, requires such significant independent creative effort that the resulting work is genuinely independent. The specification is the "idea"; the new implementation is new "expression."

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The AI Collapse of Clean Room Logic

With AI, a bad actor can take disclosed source code, prompt an LLM to "rewrite this in Python preserving all functionality," and receive — in minutes — a clean room equivalent. The AI has performed the abstraction (idea extraction) and re-expression (new implementation) simultaneously, at machine speed, with no human engineer ever reading the original code carefully enough to retain it. The clean room defense was designed to protect good-faith independent development. AI weaponizes its logic.

2.3 Derivative Works, Substantial Similarity, and Language Translation

Copyright protection extends to derivative works — works based on or derived from copyrighted originals. Translation is explicitly listed as a form of derivative work under 17 U.S.C. § 101. When a novel is translated from French to English, the translation is a derivative work requiring a license.

Can the same logic apply to code translated from TypeScript to Python to Rust? This is legally uncharted territory, and courts have not squarely addressed it. The argument for treating language translation as a derivative work is compelling: the functional architecture, the data structures, the algorithmic choices, and the overall design philosophy are all reproduced. The argument against is equally powerful: "expression" in copyright doctrine refers to the specific form of utterance, and Python and TypeScript are such fundamentally different languages — with different paradigms, different idioms, different type systems — that nothing at the expression layer is reproduced. The court would have to conclude that functional architecture itself constitutes protectable expression, which would dramatically expand copyright law's scope.

The "substantial similarity" test — the standard copyright courts apply to determine infringement — was developed with human-perceivable similarity in mind. Courts ask whether an "ordinary observer" would find the works substantially similar. An ordinary observer looking at TypeScript and Rust implementations of the same system would not find them similar at all. The test may be the wrong tool for this problem entirely.

2.4 Trade Secret Law: The Stronger Claim

Here is where the analysis shifts significantly from popular understanding. The scenario described in this paper is almost certainly not "technically legal" as commonly assumed — but the applicable law is trade secret law, not copyright law. The Defend Trade Secrets Act (DTSA, 2016) and analogous state statutes under the Uniform Trade Secrets Act (UTSA) provide

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protection for information that derives economic value from not being generally known, provided that reasonable measures are taken to keep it secret.

Proprietary source code is the paradigmatic trade secret. When that code is obtained through an unauthorized disclosure — even an accidental one through exposed source maps — and then used to create a competing product, the misappropriation claim does not evaporate because the form of the information has been changed. Courts have consistently held that trade secret protection extends to information regardless of the medium in which it is expressed. A thief who photographs a trade secret formula, then recites it from memory, then types it from recitation, does not escape liability at each transformation.

> **The Trade Secret Bridge**
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> Even where copyright fails to reach AI-assisted reimplementation, trade secret law may provide a remedy — provided the originating information was a genuine trade secret and the disclosure was unauthorized. The critical question becomes: does an accidental exposure through a debugging artifact (a source map file) constitute "misappropriation" when a third party exploits it? Courts have held that third parties who knowingly use information they have reason to know was improperly disclosed can be liable even without being the original discloser.

3. The Current Legal Landscape: Patent Law

3.1 What Patent Law Protects — And What It Does Not

Patent law, in contrast to copyright, directly protects functional innovation. A patent grants its holder the exclusive right to prevent others from making, using, selling, or importing an invention for a limited period — twenty years from the filing date in the U.S. and most jurisdictions. Unlike copyright, which arises automatically upon creation, patent protection requires affirmative application, examination, and grant.

For software, patent law is the natural home of functional protection. If Anthropic has patented specific algorithms, architectural patterns, or technical processes embodied in Claude Code, those patents would theoretically extend to functionally identical reimplementations regardless of the programming language used. A patent on a specific method of multi-turn context management in a language model runtime would be infringed by a Python reimplementation of that method just as much as by a TypeScript one.

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The problem is that patent law, as currently constituted, faces two major obstacles to addressing the AI-assisted reimplementation threat: the scope of patentable subject matter in software, and the sheer speed at which AI can generate and test variations.

3.2 The Software Patent Crisis: Alice and Its Aftermath

The Supreme Court's 2014 decision in Alice Corp. v. CLS Bank International dramatically narrowed the scope of patentable subject matter for software inventions. Under the Alice two-step test, a software patent claim is invalid if it is directed to an abstract idea and does not add "significantly more" than the abstract idea itself. The practical effect has been widespread invalidation of software patents, with courts finding that many software innovations are nothing more than abstract mathematical processes implemented on a generic computer.

The irony of Alice in the AI reimplementation context is stark. Because so many software innovations cannot survive Alice scrutiny, they are left unprotected by patent law — the one area of IP law designed to protect functional innovation. They fall back on copyright law — which, as we have seen, may not reach functional reimplementation. And trade secret law, which provides the strongest protection, is a time-limited remedy: once the secret is disclosed, the protection attenuates.

> **The Alice Trap**
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> The legal doctrine most hostile to software patents — Alice — creates an unintended vulnerability in the AI era. Innovations too abstract for patent protection, too functional for copyright protection, and too publicly disclosed for trade secret protection exist in a legal void. AI-assisted reimplementation exploits precisely this void.

3.3 The Doctrine of Equivalents: Patent Law's Closest Tool

Patent law does have a doctrine — the doctrine of equivalents — that extends patent protection beyond the literal language of claims to cover implementations that perform substantially the same function, in substantially the same way, to achieve substantially the same result. This three-part test ("function-way-result") was designed to prevent competitors from making trivial modifications to patented inventions to escape infringement.

The doctrine of equivalents is, conceptually, the most promising existing legal tool for addressing AI-assisted reimplementation. A Rust implementation of an algorithm patented in a TypeScript context performs the same function, in the same way (the same algorithmic steps), to achieve the same result (the same outputs). This should constitute infringement under the doctrine of equivalents.

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The practical obstacles are significant, however. First, the doctrine of equivalents applies only to patented inventions — requiring the hurdle of patent acquisition. Second, the "way" element of the test has been interpreted by courts to require similarity at the level of implementation mechanism, which a skilled defense attorney will argue is entirely different between TypeScript and Rust. Third, and most critically, the doctrine was designed to address human-scale trivial modifications, not AI-scale comprehensive reimplementations. Courts have never had to consider whether the doctrine applies when the "modification" was performed by a machine in hours.

3.4 The Speed Problem: Prior Art Generation by AI

Patent law rests on a fundamental social bargain: the inventor discloses her invention to the public in exchange for a limited monopoly. The disclosure serves the public good by advancing the state of the art. The monopoly serves the inventor by providing economic incentive to innovate.

AI disrupts this bargain at its foundation. If a bad actor can take any disclosed technical concept and generate dozens of functionally equivalent implementations in different languages within hours, the practical effect is to generate prior art on demand. A patent examiner reviewing a patent application has always had to search for prior art — earlier disclosures that anticipate the claimed invention. But prior art was limited by what had actually been published, filed, or publicly used before the filing date.

A sophisticated adversary with AI tools could now conceivably generate and timestamp thousands of variations of a technology before a patent application is filed, potentially defeating novelty or non-obviousness requirements. This is not a theoretical threat; it is a practical capability available today. Patent offices have not developed any framework for addressing AI-generated prior art, its authentication, or its weight in examination proceedings.

4. The Patent Cost Barrier: A System That Protects Incumbency, Not Innovation

4.1 The Structural Economics of Patent Prosecution

Patent law is theoretically neutral. It does not ask whether an applicant is a Fortune 500 corporation or a solo inventor working from a garage. It asks only whether the claimed invention is novel, non-obvious, and adequately described. In practice, however, the economics of patent prosecution impose a means test that the statute does not — and that means test has become catastrophically misaligned with the AI era's threat profile.

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A complete utility patent application, prosecuted by a registered patent attorney through issuance, costs between $15,000 and $50,000 for a moderately complex software invention — and that figure assumes a relatively smooth prosecution without multiple rounds of office actions, appeals, or inter partes review proceedings. Maintenance fees add another $10,000 to $15,000 over the patent's life. If the patent must be enforced through litigation, the costs escalate by orders of magnitude: patent litigation through trial costs an average of $3 million to $5 million per side, according to the American Intellectual Property Law Association's most recent economic survey. For a small software company or independent inventor, these figures are not merely burdensome — they are prohibitive.

The Means Test the Statute Does Not Impose

Congress intended the patent system to serve all inventors equally. The economic reality is a system where effective patent protection — the ability to obtain a patent, maintain it, and enforce it against infringement — is available as a practical matter only to entities with patent departments, litigation budgets, and the organizational staying power to survive multi-year legal proceedings. Small innovators are not merely disadvantaged; they are effectively excluded from the system's protective function.

4.2 The Timeline Paradox: Protection That Arrives After the Harm

The average time from patent application filing to issuance at the USPTO is currently 23 to 26 months for applications that receive no significant examination challenges — and many software applications face rejection rates requiring multiple rounds of response, extending prosecution to three, four, or five years. During that entire period, the applicant has "Patent Pending" status but no enforceable rights. A competitor who reads the published application — which the USPTO publishes 18 months after filing — knows exactly what is claimed and can begin designing around it, or simply copying it, with no legal consequence until issuance.

In the AI era, this timeline is not merely inconvenient — it is structurally fatal to the patent system's protective purpose for software. An AI system can replicate the functional architecture of a software product in 48 hours. The patent system's response arrives, at minimum, two years later. By the time enforceable rights exist, the harm has been complete for years, the infringing product may have captured significant market share, and the practical ability to restore the innovator's competitive position through damages and injunction is severely diminished.

The mismatch is not marginal. It is a difference of five orders of magnitude in timescale. A legal protection mechanism that operates at 730-day latency against a threat that operates at 48-hour

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latency is not providing protection — it is providing the illusion of protection while the actual harm unfolds unchecked.

4.3 The Alice Trap Amplified: Paying for Nothing

The timeline and cost problems are severe enough in isolation. Combined with the post-Alice landscape for software patent eligibility, they become devastating. As discussed in Section 3.2, the Supreme Court's Alice decision has resulted in wholesale invalidation of software patent claims that fail to demonstrate "significantly more" than an abstract idea. The practical effect is that a substantial fraction of software patent applications — estimates range from 30% to over 60% in certain technology classes — are rejected on Section 101 grounds during prosecution or invalidated post-issuance.

An independent software inventor who invests $25,000 and three years in patent prosecution has no guarantee of receiving any protection at all. If the examiner determines that the claimed innovation is an abstract idea implemented on a generic computer — a determination that is notoriously unpredictable given the lack of clear judicial guidance post-Alice — the investment is lost entirely. For a large corporation with a portfolio strategy, individual application failures are absorbed across hundreds of filings. For the independent inventor or small company, a single failed prosecution can represent an existential financial loss.

> **The Expected Value Problem**
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> When the cost of prosecution is $25,000, the probability of successful issuance is uncertain due to Alice, the time to protection is 2-4 years, and the cost of enforcement is $3-5 million, the expected value of a software patent application for a small innovator is often negative. Rational small innovators faced with these economics do not file patents — they rely on trade secrecy, speed-to-market, and copyright. All three of those alternatives are, as this paper argues, structurally inadequate against AI-enabled functional reimplementation.

4.4 AI Democratizes the Threat While Patents Remain Oligopolistic

The most damaging aspect of the patent cost barrier in the AI era is the asymmetry it creates between the threat and the defense. AI-assisted functional reimplementation is, as a capability, radically democratized. A well-resourced startup, a foreign state-sponsored entity, or a sophisticated individual actor with access to capable AI models and computing resources can replicate the functional architecture of a complex software system for a cost measured in dollars or hundreds of dollars — the cost of AI API access and engineering time.

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The defense against that threat — patent protection — remains firmly oligopolistic in its economics. The cost of offense has collapsed by orders of magnitude. The cost of defense has not moved. This asymmetry is not a market failure that will self-correct; it is a structural feature of a legal system built for a different technological era. Left unaddressed, it systematically transfers value from innovative small actors to replicating large actors — the precise inverse of the patent system's constitutional purpose.

The constitutional mandate — Article I, Section 8, Clause 8 of the U.S. Constitution — authorizes Congress to grant patents "to promote the Progress of Science and useful Arts." A patent system whose economics preclude effective participation by most inventors does not promote progress; it concentrates the benefits of innovation protection among those already most able to extract value from innovation regardless of formal legal protections. The AI era makes this constitutional drift impossible to ignore.

4.5 What Reform Must Address: The Fast-Track Registration Proposal

The reform needed is not merely fee reduction or examination acceleration, though both would help. The structural proposal this paper advances is the creation of a Fast-Track Functional Architecture Registration — a new administrative mechanism positioned between copyright registration and full patent prosecution in both cost and protection scope.

The core characteristics of this proposed registration system:

- Cost commensurate with copyright registration — in the range of $65 to $500 depending on complexity — rather than the $15,000-$50,000 of patent prosecution.
- Examination timeline of weeks rather than years — administrative review of the functional specification for clarity and completeness, not a full novelty and non-obviousness examination.
- Protection scope limited to functional architecture — the logical design, algorithmic approach, and component relationships of a software system — not the specific implementation or expression.
- Term of protection shorter than full patents — 7 to 10 years, reflecting software's faster innovation cycles — but immediately enforceable upon registration.
- Standing to seek emergency injunctive relief against AI-assisted functional reimplementation, without requiring the full evidentiary showing currently demanded before preliminary injunctions issue.
- Public registry creating constructive notice — anyone who builds a functionally equivalent system after registration date does so with notice of the registered architecture, shifting the burden of proof on independent development claims.

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This proposal does not replace the existing patent system — full patents remain available for those who want the broader protection they provide. It creates an accessible on-ramp to functional protection for the innovators the patent system currently prices out. In doing so, it realigns the economics of IP defense with the economics of AI-enabled IP offense — restoring the balance the constitutional mandate requires.

> **The Open Source Boundary**
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> A critical design question for any fast-track registration system is its interaction with open source licensing. Code released under MIT, Apache, GPL, or similar licenses carries an express grant of rights to use, modify, and redistribute. Fast-track registration should not — and constitutionally could not — override those express grants. The system proposed here targets proprietary innovations: software whose creators have chosen not to grant general use rights. Open source developers who choose to rewrite a permissively licensed codebase in a different language are doing exactly what the license authorizes. The ethical and legal distinction between that activity and exploiting a proprietary disclosure is the entire basis of this paper's argument.

5. The Ethical Dimension: Where Law Falls Short

5.1 The Law-Ethics Gap in IP

Intellectual property law has always operated with an inherent tension between legal rights and ethical norms. The law draws bright lines — something is either infringing or it is not — while ethics operates in shades of gray. In the pre-AI era, this tension was manageable because the effort required to exploit legal gray zones served as a practical deterrent. Writing a clean room reimplementation of a competitor's software in a different language required months of skilled engineering labor. The cost-benefit calculus favored legitimate competition.

AI eliminates the cost side of that equation. What was once expensive is now nearly free. What once required a team of engineers now requires a capable prompt engineer and access to a large language model. The legal gray zone that existed as a theoretical vulnerability is now an operational capability. The ethical problem — taking the fruits of another's innovation without authorization and compensation — remains exactly as severe. The legal exposure has, paradoxically, decreased because courts assessing copyright infringement look for expressive similarity that AI reimplementation deliberately eliminates.

> **The Ethical Core**
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> Ethical analysis here does not require elaborate philosophical frameworks. The question is simple: if

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> company A spends years and hundreds of millions of dollars developing a software system, and company B uses AI to recreate that system's functionality in 48 hours based on an unauthorized disclosure, has company B wronged company A? The intuitive answer is an emphatic yes. The challenge is constructing legal doctrine that reaches this obvious ethical conclusion.

5.2 Why "It's Legal" Is Not a Defense

There is a cultural pathology in the technology industry — and in legal discourse generally — that treats legality as a moral absolution. If an action is technically legal, the implicit argument goes, there is nothing ethically wrong with it. This position is philosophically untenable and historically dangerous.

Many of the most significant corporate ethical failures of the modern era involved conduct that was, at the time, arguably legal. Tax structures that stripped royalties through shell companies. Platform designs that maximized engagement while concealing harms. Data harvesting practices that disclosed themselves in terms of service no human read. In each case, "it's legal" was offered as a defense after the fact. In each case, that defense was eventually rejected — either by regulators who changed the law, or by the market, or by cultural consensus.

The AI reimplementation scenario sits precisely at this junction. The technical legality (such as it is) exists because the law has not caught up with the technology. That lag is not ethical permission. It is a window during which norms must be established — either by voluntary industry adoption or by legislative and judicial action — before the behavior becomes so entrenched that correction becomes practically impossible.

5.3 The Functional Equivalence Concept as an Ethical Standard

What ethics suggests, and what law must eventually incorporate, is a concept of functional equivalence as a harm-causing act independent of formal expression. The ethical harm is not that the Rust code looks like the TypeScript code — it does not. The ethical harm is that the Rust code does everything the TypeScript code does, for the benefit of the party that reproduced it, depriving the original creator of the economic benefit of her innovation.

Law currently recognizes functional equivalence in limited contexts. The doctrine of equivalents in patent law, as discussed, approximates this. The "look and feel" test in some copyright jurisdictions goes beyond literal copying. Trade dress protection in trademark law covers the overall visual impression of a product, not just its specific graphic elements. But no doctrine comprehensively addresses functional equivalence in software across language barriers.

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6. The AI-Specific Legal Challenges

6.1 Who Is the Infringer? Attribution in AI-Assisted Reproduction

Traditional IP infringement analysis assumes a human actor who makes deliberate choices. The person who copies a painting is the infringer. The company that manufactures a patented device is the infringer. AI complicates attribution in fundamental ways.

In the scenario under analysis, the chain of actors is: (1) whoever inadvertently exposed the source maps, (2) whoever discovered the exposure, (3) whoever forked the repository, (4) whoever prompted an AI to rewrite the code in Python, (5) the AI model itself, (6) whoever then prompted a second AI to rewrite it in Rust, and (7) whoever is exploiting the resulting code commercially. Current law has no coherent framework for distributing liability across this chain, particularly when the AI model — which performed most of the actual transformation — has no legal personhood and cannot be liable.

> **The Contributory and Vicarious Liability Question**
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> If direct liability is difficult to establish, secondary liability theories become critical. Contributory infringement requires knowledge of infringing activity and a material contribution to it. Vicarious liability requires the right and ability to control infringing activity and a direct financial benefit. Both theories were developed for human intermediaries. Applying them to AI model developers, API providers, and platform operators who have no specific knowledge of any particular infringing use will require significant doctrinal development.

6.2 The Velocity Problem: Law vs. Machine Speed

Law is a slow-moving institution by design. Legislatures deliberate. Courts hear cases over years. Appellate processes extend timelines further. The Supreme Court may hear a case on a legal question a decade after it first arises in lower courts. This deliberateness serves important values — stability, predictability, legitimacy.

AI operates at machine speed. The reimplementation scenario described in this paper — from initial disclosure to functional Rust implementation — potentially unfolds in less than 48 hours. By the time any court could issue a preliminary injunction (typically the fastest legal remedy, but still measured in weeks or months), the harm may be complete, the code may be distributed globally, and the practical ability to enforce any judgment may be substantially diminished.

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This velocity asymmetry is not merely a practical inconvenience. It represents a structural failure of law's ability to govern AI-enabled conduct. Law that operates on a timescale orders of magnitude slower than the conduct it seeks to regulate is not effectively governing that conduct at all. It is, at best, providing after-the-fact remedies for harms that have already fully materialized.

6.3 The Jurisdictional Fracture

IP law is territorial. A U.S. copyright protects against infringement in the U.S. A European patent protects against infringement in designated European jurisdictions. But AI-assisted reimplementation is jurisdictionally agnostic. Code can be generated in one country, transmitted through servers in another, stored in a third, and commercially exploited in a fourth — all within hours.

The source map exposure in the scenario here may have occurred in the United States. The forking may have occurred on servers located in multiple countries. The AI rewriting may have been performed using models hosted abroad. The resulting code may be distributed through global repositories under licenses that complicate enforcement in any single jurisdiction.

No international treaty framework addresses AI-assisted IP misappropriation comprehensively. The Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) establishes minimum standards, but those standards were negotiated in 1994 and have not been fundamentally updated to address AI. The World Intellectual Property Organization (WIPO) has convened discussions on AI and IP policy, but these discussions have not produced binding instruments.

7. Proposed Framework: Functional Equivalence as a Legal Standard

7.1 The Core Proposal: Functional Copyright

This paper proposes that intellectual property law must develop a doctrine of "functional copyright" — protection that attaches not to a specific expressive form but to the functional architecture of a software system, when that architecture represents genuine creative and innovative effort. Functional copyright would protect the logical design of a system — its data structures, algorithms, inter-component relationships, and processing pipelines — against AI-assisted reproduction regardless of the target language or implementation paradigm.

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This proposal is deliberately ambitious because incremental reform is insufficient. The existing framework — copyright protecting expression, patents protecting specific claimed inventions, trade secrets protecting confidential information — was constructed around human-scale threats. AI-scale threats require a new category.

- Functional copyright protection would require affirmative registration, similar to patent prosecution, to create notice for the public and to limit the scope of protection.
- Protection would extend for a shorter term than traditional copyright — perhaps 10 to 15 years — reflecting the faster pace of software innovation.
- The scope of protection would be defined not in terms of code but in terms of functional specifications: what the system does, the algorithmic approach it uses to do it, and the architectural relationships among its components.
- AI-generated reimplementation that reproduces a registered functional architecture would constitute infringement, regardless of the target language, provided that the functional specification of the reimplementation matches the registered specification within a defined margin.

7.2 The Functional Equivalence Test

For functional copyright, courts and administrative bodies would need a functional equivalence test — a method for determining whether two software systems are functionally equivalent for purposes of the protection. This is not a trivial problem, but neither is it an intractable one. Several approaches deserve consideration:

6.2.1 Behavioral Equivalence Testing

The most rigorous approach to functional equivalence is behavioral: two systems are functionally equivalent if they produce the same outputs for the same inputs across a comprehensive test suite. This approach is already used in software engineering for regression testing, formal verification, and competitive analysis. It could be adapted for legal proceedings, with court-appointed technical experts administering standardized test suites to the systems under comparison.

6.2.2 Architectural Fingerprinting

A less computationally intensive approach would analyze the high-level architecture of systems — the number and nature of their principal components, the data flow relationships among those components, and the algorithmic category of each component's processing. An architectural fingerprint would capture the essential design of a system in a language-agnostic representation that could be compared across implementations.

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6.2.3 AI-Assisted Equivalence Analysis

Perhaps most provocatively, AI itself could be used to assess functional equivalence. Modern large language models with code analysis capabilities can analyze two codebases and identify their functional similarities and differences with substantial accuracy. Using AI to identify AI-enabled infringement has the practical advantage of operating at machine speed — potentially enabling faster preliminary relief.

> **The Irony of Using AI to Police AI**
>
> There is something philosophically significant about deploying AI systems to detect and assess AI-assisted IP misappropriation. It suggests that AI is not merely a threat to be regulated but a tool that must be incorporated into the regulatory apparatus itself. Law in the AI age cannot afford to be AI-illiterate. Courts, patent offices, copyright registries, and trade secret arbitrators will all need AI-capable technical capacity to function effectively.

7.3 Expedited Preliminary Relief: Injunctions at Machine Speed

The velocity problem identified in Section 6.2 demands a procedural response as much as a substantive one. Courts must develop mechanisms for granting preliminary relief on timescales commensurate with the speed at which AI-enabled harms can unfold. Several reforms deserve consideration:

- Emergency TRO (Temporary Restraining Order) procedures for AI-enabled IP misappropriation, with hearing timelines measured in days rather than weeks.
- Specialized AI and IP courts or court divisions staffed by judges with technical backgrounds and AI literacy.
- Pre-registered technical specifications, held in escrow with a neutral administrative body, that can serve as immediate evidentiary foundation for emergency relief without full litigation of underlying facts.
- International mutual recognition agreements for AI-related IP emergency orders, enabling rapid cross-border enforcement.

7.4 Mandatory Provenance Tracking for AI-Generated Code

A longer-term reform would require that AI systems capable of generating substantial code maintain provenance logs — records of what inputs were used to generate what outputs, sufficient for subsequent forensic analysis. This is technically feasible: modern AI systems can

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log their inputs and outputs, and watermarking techniques for AI-generated content are an active area of research.

Mandatory provenance tracking would serve multiple purposes. It would enable tracing of AI-assisted reimplementation to its source inputs. It would create accountability for AI service providers whose systems are used for infringement. And it would shift the incentive structure: AI providers who know their systems' outputs can be traced would have strong incentives to implement safeguards against processing disclosed proprietary code.

Privacy and Competitive Concerns

Mandatory provenance tracking raises legitimate privacy and competitive concerns. Logs of AI system inputs could expose confidential business strategies, attorney-client privileged communications, or personal data. Any provenance tracking regime must be carefully designed with access controls, purpose limitations, and deletion schedules that balance IP enforcement interests against privacy interests. Treating provenance logs as analogous to metadata in litigation discovery — accessible under process but not generally public — may provide an appropriate model.

8. Can the Law Actually Change? Political Economy of IP Reform

8.1 Historical Precedents for Fundamental IP Reform

The history of intellectual property law includes episodes of fundamental reform in response to technological disruption. The development of photography prompted extension of copyright to photographic works. The invention of radio and recorded music prompted creation of new performing rights and mechanical licensing regimes. The advent of digital networks prompted the Digital Millennium Copyright Act and its international equivalents. Each of these reforms required recognition that existing doctrine was structurally inadequate and that incremental application of existing concepts could not address the new technology.

The AI challenge is larger than any of these historical precedents, but the pattern is established. Law does change in response to technological disruption. The question is not whether it can change but how much disruption must accumulate — and how much economic damage must be demonstrated — before the political will for fundamental reform crystallizes.

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8.2 The Stakeholder Landscape

IP reform is never technically neutral. Every change to IP doctrine redistributes value among stakeholders, and those stakeholders deploy substantial resources to influence the legislative and regulatory process. In the AI-era IP context, the stakeholder landscape is genuinely complex:

- Large incumbent software companies have strong interests in robust IP protection for their existing codebases, which represent decades of investment.
- AI companies have conflicting interests: they benefit from broad IP protection for their own model weights and training data, but they resist constraints that would limit their systems' ability to process and generate code.
- Open source communities have historically opposed software patents and have complex positions on copyright — favoring copyleft licenses that use copyright law instrumentally to mandate openness.
- Academic and research institutions favor broad access to technical knowledge and resist IP regimes that impede research.
- Nations with large software export industries (U.S., EU, India, China) have different positions based on their comparative advantage in innovation versus implementation.

Reform that meaningfully addresses AI-enabled IP misappropriation will require sufficient alignment among these competing interests to achieve legislative majority — a high bar in any jurisdiction, and an extraordinarily high bar internationally. The more likely near-term path is judicial development of existing doctrines, supplemented by administrative guidance from copyright offices and patent offices, while legislative reform develops more slowly.

8.3 The International Coordination Problem

Because AI-assisted reimplementation is inherently cross-border, effective reform requires international coordination. National IP law that robustly protects against functional equivalence is defeated if the reimplementation can be performed in a non-participating jurisdiction. This creates a race-to-the-bottom dynamic in which jurisdictions with weaker IP protection attract the most commercially disruptive AI operations.

International IP coordination historically requires extended multilateral negotiation. The TRIPS Agreement took years to negotiate and remains contested. A new instrument addressing AI and IP would face similar obstacles, complicated by the fact that countries have sharply divergent interests in AI development and deployment. The United States and European Union are, broadly, net exporters of AI-protectable innovation; China and other emerging economies are

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rapidly developing competitive AI capacity and may resist regimes that protect existing incumbents.

A more realistic near-term approach might be bilateral or plurilateral agreements among jurisdictions with aligned interests — potentially built on the foundation of existing trade agreements — establishing mutual recognition of AI-era IP protections. The U.S.-EU Trade and Technology Council provides one possible vehicle. Regional agreements like the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) provide another.

9. Technical Dimensions: What the Law Must Understand

9.1 Source Maps: A Technical Vulnerability and Legal Artifact

The specific vulnerability exploited in this scenario — source map files — deserves technical explanation, because courts and legislators who do not understand the technical context cannot craft appropriate doctrine.

Source maps are files generated by build tooling (compilers, bundlers, transpilers) that map positions in compiled or minified output code back to positions in the original source code. They are essential for debugging: when a compiled application throws an error, the developer's tools use the source map to show the error in terms of the original source rather than the incomprehensible compiled output. Source maps are typically generated during the build process and deployed alongside production code, but are intended to be accessible only to developers with access to the development tools.

When source maps are inadvertently exposed to the public internet — for example, when a deployment pipeline fails to restrict access to .map files — the original source code (or a very close approximation of it) becomes accessible to anyone who knows to look for it. The source map contains the full source text of the original files, function names, variable names, and line-level mappings. It is, in effect, the source code with additional metadata.

> **Legal Significance of Source Map Exposure**
>
> Source map files are almost certainly trade secret information when associated with proprietary software. Their inadvertent exposure likely does not constitute publication or dedication to the public domain under copyright doctrine. A party who discovers exposed source maps and deliberately uses them to reconstruct proprietary code has almost certainly committed trade secret misappropriation, regardless of whether they have committed copyright infringement. The deliberateness element —

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> knowing that the information is proprietary and using it anyway — is key to the misappropriation analysis.

9.2 The Language Translation Capability: What AI Actually Does

To assess the legal and policy implications of AI-assisted code translation, it is essential to understand what modern large language models actually do when asked to translate code between programming languages.

Modern LLMs trained on large code corpora have internalized not just the syntax of multiple programming languages but the idiomatic patterns, standard libraries, design patterns, and common algorithms of each. When asked to translate a TypeScript codebase to Python, a capable model does not perform mechanical syntax substitution. It understands the TypeScript code's functional intent — what each function does, what data structures represent, how components interact — and then produces Python code that implements the same functional intent using Python idioms and patterns.

The result is code that may look almost nothing like the original at the surface level. Variable names are changed to Python convention. Class hierarchies are restructured to use Python's duck typing rather than TypeScript interfaces. Asynchronous patterns shift from TypeScript promises to Python asyncio. Error handling idioms differ. But the functional behavior — what the software does — is reproduced with high fidelity.

This is not mechanical copying. It is something closer to comprehension and re-expression. The AI has, in a meaningful sense, understood the original and produced an independent expression of the same understanding. Patent law, as discussed, has tools to address this — the doctrine of equivalents. Copyright law does not, and that is the structural problem.

9.3 The Rust Rewrite: An Additional Layer

The scenario adds a further step: after the Python reimplementation, a subsequent AI-assisted rewrite produces a Rust implementation. This additional step is legally significant because it puts even more distance between the original TypeScript source and the final artifact. Rust's ownership model, lifetime system, and zero-cost abstractions produce code that looks radically different from both TypeScript and Python implementations of the same functionality.

The practical effect is that a substantial similarity analysis — even one extended to consider architectural features — becomes progressively harder to sustain as the chain of

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transformations lengthens. Each step adds a layer of genuine transformation: Python's dynamic typing and garbage collection differ fundamentally from TypeScript's type system; Rust's ownership model differs fundamentally from both. At each step, the argument becomes easier to make that the new implementation is independent, even though the functional heritage is unbroken.

This progressive distancing through chained AI transformation may become a deliberate evasion strategy. Bad actors who understand IP doctrine could design workflows specifically to maximize the distance from any original while preserving functional equivalence — a form of legal arbitrage that existing law cannot prevent.

10. The Compiled Binary Problem: The Perfect Evidentiary Void

The analysis in prior sections contains an assumption so fundamental it went unstated — and that assumption is wrong. The functional equivalence testing, architectural fingerprinting, behavioral comparison, and AI-assisted similarity analysis proposed in Section 7 all rest on one premise: that someone can examine the allegedly infringing implementation. In the real world of compiled software, that premise fails catastrophically.

When Rust source code is compiled to a native binary — or when Go, C, or C++ code is compiled — the resulting artifact is machine code: a sequence of processor instructions with no meaningful relationship to the source that produced it. Function names become memory addresses. Variable names disappear entirely. Data structures are flattened into memory layouts. The elegant architectural decisions, the algorithmic choices, the component relationships that constituted the "functional expression" we sought to protect — all of it is gone, rendered into an opaque binary blob that reveals nothing about its origin.

> **The Enforcement Void**
>
> Every legal theory discussed in this paper — copyright infringement, trade secret misappropriation, patent infringement under the doctrine of equivalents, functional copyright — requires some ability to examine and compare implementations. A compiled binary without source code access is, for practical enforcement purposes, a black box. The bad actor who ships only a compiled Rust binary derived from leaked proprietary TypeScript has, in a very real sense, laundered the evidence of infringement out of existence.

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10.1 What Compilation Does — and Does Not — Hide

To be precise about the problem, it is worth distinguishing what compilation actually conceals versus what forensic tools can sometimes recover.

10.1.1 What Is Effectively Concealed

- All source-level identifiers: function names, variable names, class names, module names, comments, documentation strings
- All architectural organization: file structure, module boundaries, package hierarchy, namespace organization
- All language-specific idioms: the patterns that identify a codebase as idiomatic TypeScript, Python, or Rust
- The development history: commit messages, author information, the incremental steps by which the code was produced
- Any AI prompts or intermediate artifacts used during the reimplementation process, if not separately logged

10.1.2 What May Survive — With Significant Effort

- Broad algorithmic structure: sophisticated reverse engineers can sometimes identify well-known algorithms (sorting routines, cryptographic primitives, compression schemes) from their binary signatures
- Numerical constants and magic values: specific constants embedded in code (hash seeds, buffer sizes, protocol identifiers) sometimes survive compilation and can serve as fingerprints
- Control flow graphs: the high-level structure of branches, loops, and function calls can be partially reconstructed by decompilers, though with enormous ambiguity
- String literals: human-readable strings embedded in binaries (error messages, configuration keys, log output) are not compiled away and can reveal functional heritage

The partial survivability of some information through compilation is the basis for binary forensics — a specialized field used in malware analysis, vulnerability research, and, occasionally, IP litigation. But the evidentiary standard for IP infringement is not "a forensic expert can find hints of similarity." It is demonstrable copying or functional reproduction, proven by a preponderance of the evidence. Binary forensics, even at its most sophisticated, rarely rises to that standard against a determined adversary.

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10.2 The Money Laundering Analogy: Structured Transformation as Evasion

The chained AI transformation scenario — TypeScript leaked code → AI rewrite to Python → AI rewrite to Rust → compiled binary → shipped product — is structurally identical to money laundering through layered transactions. In financial crime, "layering" is the process of conducting multiple transactions to obscure the original source of funds. Each transaction makes tracing more difficult. Enough layers make tracing practically impossible.

IP laundering through chained AI transformation operates on the same logic. Each transformation step — each language change, each idiom shift, each compilation — adds a layer of distance from the original. A sophisticated bad actor does not need to understand IP law in detail. They only need to understand that more transformation steps equals less evidentiary exposure. The playbook writes itself:

- Obtain leaked source code through any means — exploit exposed source maps, use a disposable account, access through a trusted insider
- Feed it to an AI in a jurisdiction with weak IP enforcement or through an anonymizing service with no logging
- Generate Python first — superficially different, same function
- Generate Rust from the Python — now two steps removed from the original
- Compile to a native binary — the source is gone
- Strip debug symbols from the binary — remove even the residual naming information
- Ship the binary — legally, the only artifact that exists in the world is machine code with no visible lineage.

11. The Governance Gap: When Law Cannot Move at Machine Speed

11.1 The False Premise: Law as the Primary Governance Mechanism

The preceding sections have argued for legal reforms — functional copyright, fast-track registration, expedited injunctive relief, mandatory provenance tracking. Those arguments are sound. But they rest on an assumption that must be confronted directly: that law is the primary mechanism through which the harms described in this paper will be governed. That assumption is wrong, and the paper's policy recommendations are incomplete without acknowledging it.

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Law is one governance mechanism among several. It is the slowest, the most expensive to invoke, and the most dependent on cooperative actors amenable to jurisdiction. In the AI-enabled IP misappropriation context, law will almost always arrive after the harm has fully materialized. The question is not how to make law fast enough to prevent the harm — it cannot be made fast enough. The question is what fills the governance gap while law develops, and what law's role should be understood to be once the gap is acknowledged honestly.

> **Law as Ratification, Not Prevention**
>
> Historical analysis of technology disruption and legal response reveals a consistent pattern: technology creates the practice, leading actors establish norms, and law eventually codifies and universalizes those norms. The DMCA's notice-and-takedown system ratified what platforms were already doing voluntarily. GDPR codified privacy practices that market leaders like Apple had already adopted as competitive differentiation. The lesson is that law's realistic role in AI-era IP governance is norm ratification and universal enforcement — not prevention of first-mover harm. The urgency, therefore, is in establishing norms now, before bad practices become so entrenched that even ratification becomes politically impossible.

11.2 The Non-Law Governance Toolkit

Several governance mechanisms operate at technology speed rather than legislative speed, and each has a role to play in the interim period before legal reform matures:

11.2.1 Platform Governance: The Chokepoint Strategy

The distribution of software — even compiled binaries — requires infrastructure: code repositories, package registries, cloud hosting, content delivery networks. GitHub, npm, PyPI, Hugging Face, AWS, Cloudflare, and their peers are chokepoints through which virtually all software distribution flows. These platforms already enforce terms of service that prohibit certain categories of content — they remove CSAM, honor DMCA takedown notices, and implement sanctions compliance — at machine speed, without waiting for court orders.

Platform governance could extend to AI-enabled IP misappropriation through expanded terms of service prohibiting the distribution of software known or reasonably suspected to be derived from unauthorized disclosure of proprietary code, combined with automated detection systems and rapid takedown procedures. This is not a perfect solution — determined actors can use lesser-known hosting infrastructure — but it raises the cost and friction of misappropriation significantly, which is often sufficient to deter purely commercial actors.

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11.2.2 AI Provider Governance: Filtering at the Point of Generation

The AI systems capable of performing the language-to-language functional reimplementation described in this paper are provided by a relatively small number of companies — Anthropic, OpenAI, Google, Meta, Mistral, and a handful of others. These providers can implement governance unilaterally through their acceptable use policies: prohibiting the use of their systems to process disclosed proprietary code for reimplementation purposes, implementing input screening, and maintaining provenance logs.

This is not a hypothetical capability. AI providers already decline certain categories of requests — assistance with weapons development, generation of CSAM, specific categories of fraud facilitation. Adding IP misappropriation to that list is a policy decision, not a technical limitation. The challenge is enforcement: users who intend to misappropriate can obfuscate their inputs, use open-source models without API restrictions, or access models through jurisdictions with no acceptable use enforcement. But governance at the provider level, combined with platform governance at the distribution level, creates overlapping friction that is meaningfully deterrent against commercial actors.

11.2.3 Technical Standards: Provenance Watermarking

Standards bodies — IEEE, IETF, NIST — move faster than legislatures. A technical standard for AI-generated code provenance watermarking — embedding cryptographically verifiable metadata in AI-generated code that survives language translation and even compilation to some degree — could be drafted, reviewed, and adopted within 18 to 24 months. Several research groups are actively working on watermarking techniques for AI-generated content. A coordinated standards effort, potentially supported by NIST's AI Risk Management Framework infrastructure, could accelerate this significantly.

Provenance watermarking does not prevent misappropriation. It creates a forensic trail that survives the transformation chain, making detection possible even after binary compilation. Combined with platform governance that checks for watermark presence, it creates a system in which AI-assisted reimplementation using reputable providers leaves detectable evidence — shifting the evidentiary burden in subsequent legal proceedings and deterring actors who cannot afford forensic exposure.

11.3 The Barn Is Already Open: The Horses That Do Not Come Back

There is a point in any policy analysis where intellectual honesty requires confronting the hardest version of the problem. For AI-enabled IP misappropriation, that hardest version is this: for a window of time — likely five to ten years — determined actors will be able to execute the

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misappropriation playbook faster than any governance mechanism, legal or otherwise, can respond. Some harms will not be prevented. Some companies will be damaged or destroyed before any remedy is available.

The conventional policy response to this acknowledgment is to minimize it — to argue that even imperfect deterrence is valuable, that after-the-fact damages provide some compensation, that the legal framework being built today will prevent future harms even if it cannot address current ones. All of these arguments are true. None of them are adequate.

When a small innovative company is driven out of business by AI-assisted competitive misappropriation — when its revenue collapses faster than its legal remedies can materialize, when its engineers disperse, when its investors write off their positions — the harm is not merely to that company. The harm is to the innovation ecosystem from which the next generation of technology would have grown. The people who would have founded the company's successor, who would have built on its technical foundation, who would have attracted capital based on its demonstrated market validation — they are dispersed, their trajectory altered, their contribution to the innovation ecosystem permanently diminished.

The Irreversibility Problem

Legal systems can award damages. They cannot reconstitute dissolved companies, reassemble dispersed engineering teams, restore lost market positions, or compensate for innovations that were never built because the company that would have built them no longer exists. The irreversibility of AI-enabled competitive destruction is the most compelling argument for prophylactic governance — platform controls, provider restrictions, provenance standards — that operate before harm materializes rather than after.

11.4 The China Parallel: State-Sponsored Economic Warfare at AI Speed

The scenario analyzed throughout this paper — AI-assisted functional reimplementation of disclosed proprietary code — is most alarming not as a commercial threat but as a state-sponsored economic warfare capability. To understand why, it is necessary to examine what state-sponsored IP theft has already accomplished on a slower timescale, and then project what AI acceleration does to that calculus.

The Commission on the Theft of American Intellectual Property — a bipartisan body co-chaired by former Ambassador Jon Huntsman and former Director of National Intelligence Admiral Dennis Blair — estimated in its 2017 report that Chinese theft of American IP costs between $225 billion and $600 billion annually. Even accepting the methodological criticisms of the high

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end of that range, the scale is extraordinary: a sustained, deliberate transfer of innovation value from American companies to Chinese state-backed competitors, executed over decades through cyber espionage, forced technology transfer requirements, and systematic appropriation of trade secrets.

The mechanism of China's long-game IP theft followed a recognizable pattern across multiple industries: identify strategic technology sectors aligned with Made in China 2025 objectives; acquire the technology through whatever means were available — corporate espionage, insider recruitment, forced joint ventures, cyber intrusion; establish domestic manufacturers with access to the acquired technology, subsidized by state capital and shielded from competitive pressure by market access restrictions; and systematically undercut the original innovators on price until they lost market share, lost funding, and exited the market or were acquired at distressed valuations.

American solar panel manufacturers, advanced materials companies, telecommunications equipment producers, and semiconductor firms all experienced versions of this playbook. The damage was not merely commercial. It was structural: entire US industrial capabilities in strategic sectors were eliminated or severely diminished. The jobs were not just lost — the institutional knowledge, the engineering talent pipelines, the supplier ecosystems, and the accumulated technical expertise represented by those companies were dispersed and in many cases permanently lost to the US economy.

> **The Acceleration Factor**
>
> China's multi-decade IP theft campaign required years to execute at each stage: years to acquire technology, years to build manufacturing capacity, years to achieve competitive scale, years for the market impact to materialize. That temporal extent created windows for detection, diplomatic response, trade sanctions, and policy adjustment — windows that, while often missed or ignored, at least theoretically existed. AI eliminates those windows. The acquisition stage collapses from years to hours. The reimplementation stage collapses from months to days. The deployment stage is limited only by cloud infrastructure provisioning time. The entire campaign that took China a decade can now be executed in a quarter.

11.5 Two Threat Models, Two Response Frameworks

The central policy error that this paper warns against is treating AI-enabled IP misappropriation as a single, unified threat amenable to a single response framework. It is not. There are two distinct threat models, each requiring a fundamentally different response toolkit, and conflating them leads to under-response on both fronts.

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11.5.1 Threat Model A: Commercial Misappropriation

The commercial misappropriation threat involves private actors — well-funded startups, established competitors, criminal organizations — motivated by competitive advantage or profit. These actors are, in principle, subject to legal jurisdiction, deterred by liability risk, and constrained by reputational considerations. For this threat model, the legal and governance reforms proposed in earlier sections of this paper are the appropriate response: functional copyright doctrine, fast-track registration, expedited injunctive relief, platform governance, AI provider restrictions, and provenance standards.

The realistic expectation for commercial misappropriation is not zero harm but managed harm — a governance environment in which the cost and friction of misappropriation are high enough to deter rational commercial actors, in which detection is sufficiently likely to create liability exposure, and in which remedies, while imperfect, are available with sufficient speed to provide meaningful deterrence. This is the same standard to which we hold corporate fraud, financial crime, and environmental violation enforcement — not perfect prevention, but credible deterrence through a combination of ex ante governance and ex post liability.

11.5.2 Threat Model B: State-Sponsored Economic Warfare

The state-sponsored threat model is categorically different in ways that make the entire legal reform framework proposed in this paper essentially irrelevant as a primary response. State actors operating from non-treaty jurisdictions are not subject to US court orders. Their assets are sovereign and cannot be seized to satisfy judgments. They are not deterred by commercial liability — they are executing geopolitical strategy with state resources, and a damage judgment in a US court is a domestic political irrelevancy, not a business constraint.

The appropriate response toolkit for state-sponsored AI-enabled IP theft operates entirely outside IP law. It encompasses export controls administered by the Bureau of Industry and Security under the Export Administration Regulations — the same framework currently being deployed against advanced semiconductor technology transfer to China. It encompasses CFIUS review of foreign investment in US technology companies. It encompasses USTR Section 301 investigations and retaliatory trade measures. It encompasses counterintelligence operations and intelligence community attribution capabilities. And it encompasses diplomatic pressure through bilateral and multilateral channels — the same mechanisms that produced the 2015 Obama-Xi agreement nominally renouncing commercial cyber espionage.

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The existing export control framework has already begun to grapple with AI-specific threats. The Bureau of Industry and Security has implemented controls on advanced computing integrated circuits and, in January 2025, issued rules controlling AI model weights — the trained parameters of advanced AI systems — as export-controlled items. These controls represent a recognition that the AI technology stack, not just the underlying hardware, has national security implications that require active management rather than open market distribution.

> **The Export Control Gap for Software Reimplementation**
>
> The existing export control framework addresses the export of hardware and model weights. It does not yet address the use of AI systems to perform functional reimplementation of controlled technology — the use of an AI model, potentially operating on US infrastructure, to reproduce the functional architecture of a US proprietary software system for the benefit of a foreign state-sponsored actor. This is the gap that AI-enabled IP theft exploits in the national security context, and it requires explicit regulatory attention under the Export Administration Regulations or the International Emergency Economic Powers Act framework.

11.6 The Legislative Vehicles: Where Hooks Already Exist

While comprehensive legislative reform of IP law to address AI-enabled misappropriation will take years, several existing and proposed legislative vehicles contain hooks that can be used or expanded to address specific aspects of the threat in the near term:

11.6.1 Export Administration Regulations and IEEPA

The Export Administration Regulations, administered by BIS under the authority of the Export Control Reform Act and the International Emergency Economic Powers Act (IEEPA), already provide a framework for restricting the transfer of technology with national security implications. Executive Order authority under IEEPA has been used to impose investment restrictions, technology controls, and entity-specific sanctions on AI-related timescales measured in months. An executive order specifically addressing AI-assisted functional reimplementation of US-origin software technology in the state-sponsored threat context could be implemented without new legislation, building on the framework already established for hardware export controls.

11.6.2 The GAIN AI Act and AI Chip Governance

The Guaranteeing Access and Innovation for National Artificial Intelligence Act, under consideration in the House of Representatives as of early 2026, would impose restrictions on AI chip exports prioritizing US customers before export to countries of concern. While primarily focused on hardware, the GAIN AI Act represents legislative momentum toward a

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comprehensive AI national security framework. Amendments addressing the software and functional reimplementation dimensions of AI-enabled IP theft could be incorporated into this legislative vehicle without requiring a standalone bill — a significant advantage given congressional bandwidth constraints.

11.6.3 The Remote Access Security Act

The Remote Access Security Act, passed by the House of Representatives in January 2026, would expand US export controls to cover the provision of cloud-based services — addressing the gap through which adversaries access US AI infrastructure without importing controlled hardware. This is directly relevant to AI-enabled reimplementation: a foreign state-sponsored actor that accesses a US AI API to perform functional reimplementation of US proprietary software is engaging in the precise category of cloud-based technology transfer that the Remote Access Security Act targets. Explicit regulatory guidance under this framework, specifying that AI-assisted functional reimplementation constitutes a controlled transfer, would extend the export control framework to the software reimplementation threat.

11.6.4 CFIUS and Strategic Technology Protection

The Committee on Foreign Investment in the United States review process, significantly expanded by the Foreign Investment Risk Review Modernization Act of 2018, provides a mechanism for blocking foreign acquisition of US companies in critical technology sectors. CFIUS review is triggered by foreign investment in covered businesses — including those developing critical technologies. Regulatory guidance expanding CFIUS jurisdiction to include foreign-funded ventures that are commercially exploiting AI-assisted reimplementation of US proprietary technology would create a merger-control backstop against the most organized and well-capitalized forms of state-sponsored misappropriation.

11.6.5 The Economic Espionage Act

The Economic Espionage Act of 1996 criminalizes the theft of trade secrets for the benefit of a foreign government, instrumentality, or agent — carrying penalties of up to 15 years imprisonment and fines of up to $5 million for individuals, and fines of up to $10 million for organizations. AI-assisted functional reimplementation of US proprietary software at the direction or for the benefit of a foreign state actor constitutes economic espionage under existing statutory language. What is missing is not the legal authority but the investigative priority and technical capability to detect, attribute, and prosecute AI-enabled economic espionage before the evidence disappears into compiled binaries and dispersed infrastructure. Dedicated FBI and DOJ resources specifically targeting AI-enabled economic espionage, with

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the technical forensic capabilities necessary to trace provenance chains, would extend existing statutory authority to the new threat.

11.7 The Honest Assessment: What Is and Is Not Achievable

Intellectual honesty demands a clear statement of what this paper's proposed reforms can and cannot achieve:

- Reformed IP law — functional copyright, fast-track registration, expedited injunctions — can deter and provide remedies against rational commercial actors operating within treaty-bound jurisdictions. It cannot prevent all commercial misappropriation and cannot reach state-sponsored actors.
- Platform and AI provider governance can raise the friction cost of misappropriation against actors dependent on mainstream infrastructure. It cannot prevent misappropriation by actors with access to alternative infrastructure or open-source models.
- Export control expansion under IEEPA and EAR can address state-sponsored AI-enabled IP theft as a national security matter, operating on timescales of months rather than years. It cannot address actors who have already built domestic AI capability independent of US infrastructure.
- Provenance watermarking standards can create forensic trails that survive the transformation chain and support detection and attribution. They cannot prevent misappropriation by actors who use non-watermarking-compliant systems.
- No governance framework can prevent all harms during the transition period. Some innovative companies will be damaged or destroyed before any remedy is available. The policy goal is not to prevent all harm but to establish the framework within which harms become progressively less frequent and remedies progressively more available.

The parallel to environmental regulation is apt and instructive. The Clean Air Act of 1970 did not immediately stop industrial air pollution. It established a framework within which enforcement became progressively more effective over decades, technology improved, and the worst harms were eventually brought under meaningful control. The innovators damaged by pollution in 1972 received inadequate remedy. The framework they helped build protected those who came after. That is the realistic aspiration for AI-era IP governance: not justice for the first victims, but a framework that prevents the last ones.

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12. The Geopolitical Equalizer: AI Economic Warfare, Small State Power Projection, and the Collapse of the Military-Economic Correlation

12.1 The Question Nobody in Policy Will Say Out Loud

This paper began as an analysis of intellectual property law. It has arrived, by logical progression, at a question that IP law cannot answer and that the policy establishment is deeply reluctant to ask: at what point does systematic, AI-accelerated economic destruction constitute an act of war — and what are nations entitled to do in response?

The reluctance to ask this question is understandable. It leads to uncomfortable places. It implicates military doctrine, nuclear deterrence theory, international humanitarian law, and the foundational architecture of the post-WWII international order. Legal scholars prefer to stay within their lanes. IP lawyers analyze IP law. Military strategists analyze military strategy. The question you are now reading exists in the gap between those lanes — and that gap is precisely where AI-enabled economic warfare is being waged, deliberately and successfully, by adversaries who understand that the gap offers unlimited offense with constrained response.

> **The Strategic Exploitation of Legal Ambiguity**
>
> The absence of a defined legal and strategic framework for responding to AI-accelerated economic warfare is not a neutral condition. It is a one-sided advantage for aggressors. An actor who knows their economic warfare will never be recognized as armed aggression faces no deterrence constraint from that recognition. They can escalate indefinitely, knowing that their victim's response options are constrained to diplomatic protest, sanctions, and civil litigation — none of which operate at the speed or scale necessary to prevent the harm.

12.2 The Power Asymmetry Reversal — What AI Actually Changes

Military power has historically required a convergence of prerequisites that correlated tightly with national size and wealth: population for manpower, industrial base for weapons production, sustained economic output for logistics, geographic strategic depth, alliance relationships, and massive capital investment in platforms — carriers, aircraft, armored formations, missile systems. These prerequisites created a relatively stable international hierarchy in which small, poor states were fundamentally constrained in their ability to threaten large, wealthy ones.

AI severs every single one of those correlations simultaneously. Consider what a capable AI-enabled economic warfare operation actually requires: a small number of elite technical

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operators — dozens, not millions; computing infrastructure rentable from cloud providers or buildable domestically for tens of millions rather than the trillions that a carrier battle group costs; AI models increasingly available as open source requiring no supply chain; network access available globally; and a target — any nation whose companies have digital infrastructure.

This is not a theoretical future capability. It is a present operational reality. North Korea's Lazarus Group — a state-sponsored operation assessed by the FBI, Chainalysis, and multiple intelligence agencies as controlled by Pyongyang's Reconnaissance General Bureau — has demonstrated exactly what AI-amplified economic warfare looks like at the pre-AI frontier. The numbers are extraordinary in their strategic implication.

Between 2021 and 2025, the Lazarus Group stole over $6.75 billion in cryptocurrency — with $2.02 billion stolen in 2025 alone, a 51% increase year-over-year, representing approximately 60% of all global cryptocurrency theft in that year. The February 2025 Bybit exchange compromise alone yielded $1.5 billion in Ethereum — the largest single financial theft in recorded history — executed through exploitation of a supply chain vulnerability in transaction signing software. Within 48 hours, $160 million had been laundered through mixer services, chain-hopping, and OTC brokers — a speed of financial obfuscation that, according to blockchain analytics firm TRM Labs, would have been unimaginable a year earlier.

The Strategic Arithmetic of North Korea's Cyber Operation

North Korea's GDP is estimated at approximately $18 billion annually. The Lazarus Group's 2025 cryptocurrency theft alone — $2.02 billion — represents roughly 11% of the entire national economy generated through a handful of technical operators, no conventional military engagement, and no supply chain exposure. A UN Security Council sanctions committee report found that North Korea's cyberattacks generate approximately 50% of the country's foreign currency income and fund an estimated 40% of its weapons of mass destruction programs. The sanctions regime the international community spent decades constructing has been partially offset by a capability that requires a fraction of 1% of the resources that regime was designed to deny.

12.3 The AI Leap: From Cyber Theft to Comprehensive Economic Warfare

What North Korea has demonstrated with pre-AI cyber capabilities is the proof of concept. What AI enables is orders of magnitude beyond that proof of concept — not in degree but in kind. The pre-AI Lazarus Group required extraordinarily skilled human operators, years of training pipelines, manual exploitation of specific vulnerabilities, and human-speed operations limited by operator cognitive capacity. AI changes every one of those constraints.

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AI multiplies operator leverage by orders of magnitude, automating vulnerability discovery, exploitation, lateral movement, and evidence destruction simultaneously. It reduces the skill floor dramatically — operators direct AI rather than manually executing complex attack chains. It enables simultaneous operations at scale that is simply impossible for human teams. And critically for the IP theft context this paper has analyzed throughout: it enables the specific capability of functional reimplementation of proprietary technology at machine speed, with automated evidence elimination through compilation and symbol stripping.

The resulting weapons taxonomy extends well beyond cryptocurrency theft or IP reimplementation. AI-enabled economic warfare encompasses at minimum five distinct weapon categories, each deployable below the threshold of recognized armed conflict:

- Automated IP extraction and reimplementation at scale — not a single incident but a continuous automated operation systematically stripping strategic software sectors of their proprietary value, delivering functional equivalents to state-directed enterprises on a timescale of weeks rather than years.
- AI-enabled financial market warfare — automated trading systems combined with AI-generated disinformation can move markets with precision. The 2010 Flash Crash, caused by algorithmic trading without deliberate AI coordination, erased approximately $1 trillion in market capitalization in minutes before partial recovery. A deliberately engineered version with AI amplification and no recovery intent is a precision economic weapon.
- Supply chain infiltration through AI-generated compromised code — AI can generate subtly backdoored code that passes automated security review and careful human review, inserted into open source dependencies used by critical infrastructure, financial systems, or defense contractors. The SolarWinds attack — manual, eventually detected — previews what an AI-automated version at continental scale looks like.
- AI-accelerated social and economic destabilization — synthetic media, AI-generated disinformation at massive scale, targeted manipulation of financial sentiment, automated amplification of social division — all operating below the threshold of recognized armed conflict, all causing measurable economic damage.
- Autonomous economic attrition — the combination of all four preceding capabilities operating simultaneously against a target economy, no single operation crossing the threshold of recognized aggression, the cumulative effect equivalent to a sustained bombing campaign against economic infrastructure.

The strategic significance of this taxonomy is that each weapon is individually deniable, individually below the recognized threshold of armed conflict, and individually addressable

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through existing civil and criminal law mechanisms — slowly, expensively, and often after the harm is complete. The combination, operated simultaneously by a state actor with AI amplification, constitutes comprehensive economic warfare that existing frameworks are structurally unable to recognize as such.

12.4 Unrestricted Warfare: The Doctrine That Predicted This

In February 1999, two senior colonels of the People's Liberation Army Air Force — Qiao Liang and Wang Xiangsui — published a strategic treatise that has since become required reading at West Point. Titled "Unrestricted Warfare" (超限战, literally "warfare beyond bounds"), it argued that the primary weakness of the United States in military matters was its fixation on technology and kinetic force as the definition of warfare — and that this blind spot made the US comprehensively vulnerable to attack along asymmetric dimensions it did not recognize as warfare at all.

The book's central thesis was that a technologically inferior nation could defeat a militarily superior opponent by conducting warfare across every domain simultaneously — legal, economic, informational, financial, cyber, psychological — while carefully remaining below the threshold that would trigger the superior power's recognized response frameworks. The first rule of unrestricted warfare, Qiao stated in subsequent interviews, is that there are no rules. The objective is not to win a battle but to win the war by making the adversary incapable of sustaining one.

AI transforms unrestricted warfare from a theoretical framework requiring patient human execution across decades into an operational capability executable at machine speed. What Qiao and Wang described as a slow but inexorable assault requiring years of deliberate human effort can now be compressed into months or weeks through AI automation. The doctrine has found its weapon.

> **The Doctrine Applied to AI-Enabled IP Warfare**
>
> Unrestricted Warfare specifically identified attacks on financial systems, theft of intellectual property, and exploitation of legal frameworks as legitimate instruments of warfare against a technologically superior adversary. AI-enabled functional reimplementation of proprietary software — conducted below the threshold of recognized IP infringement through compiled binary distribution, executed across jurisdictional boundaries, exploiting the velocity gap between AI-speed harm and legal-speed remedy — is unrestricted warfare implemented with precision instruments the doctrine's authors could not have imagined in 1999.

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12.5 The Colt Revolver Moment: AI as the Geopolitical Equalizer

The Colt revolver was marketed in the 1830s as "the equalizer" — a weapon that allowed a physically smaller person to defend against a larger one by negating the advantage of size and strength. The same logic has appeared repeatedly in military history: the longbow at Agincourt, the machine gun in the trenches, the submarine in the Atlantic, the improvised explosive device in Iraq. Each represented a technology that partially negated an existing power advantage, allowing smaller or weaker actors to impose costs on larger, stronger ones.

AI is the geopolitical equalizer for the economic warfare domain — not a partial negation of existing power advantages but a comprehensive one. For the first time in the history of state competition, a nation with 25 million people and a $20 billion GDP, unable to field a competitive conventional military, unable to build a blue-water navy, unable to project air power beyond its borders, can operate an economic warfare capability that causes damage to a $25 trillion economy measurable in hundreds of billions annually.

This is not a projected future. North Korea has already demonstrated it at the pre-AI frontier. What AI enables is that same capability at dramatically lower cost, dramatically higher scale, and dramatically lower skill requirements — extending it from the handful of nations with North Korea's extraordinary cyber investment to any state-level actor with the political will to deploy it and the computing infrastructure to support it. The bar for entry into devastating economic warfare has collapsed.

The historical precedents for this kind of power equalization are not comforting. Every previous technology that significantly equalized military power — from gunpowder to nuclear weapons — produced a period of strategic instability, miscalculation risk, and often conflict before the international order adapted to incorporate the new capability into its deterrence frameworks. Nuclear weapons produced four decades of Cold War and multiple near-misses before a stable deterrence architecture emerged. AI-enabled economic warfare is potentially more destabilizing because it lacks the visibility, the attribution clarity, and the defined threshold that made nuclear deterrence, however precarious, ultimately workable.

12.6 The Existential Equivalence Question

We arrive at the question your paper must ask directly: how is AI-accelerated economic warfare existentially different from war over oil or water rights?

International law and strategic doctrine have long recognized that nations may use military force — including preemptive force under the Caroline doctrine — to defend against existential threats to their capacity to sustain and defend themselves. Wars have been fought, and legally

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recognized as justifiable, over oil supply (the Gulf War), water access (multiple Middle Eastern conflicts, projected future conflicts over the Nile and Himalayan glacier melt), and territory that controls trade routes or produces food. The legal and moral justification rests on a single foundation: the threat is existential to the nation's capacity to sustain itself and its people.

Now apply that framework to AI-accelerated IP theft and economic warfare at scale. Systematic destruction of a nation's innovative capacity destroys its ability to generate wealth in the sectors that matter most to future economic and military power. Loss of strategic technology sectors eliminates military capability over time — you cannot build next-generation weapons systems if you have lost your semiconductor industry, your advanced software engineering talent base, your AI research ecosystem. Loss of economic leadership translates directly and historically into loss of geopolitical power, loss of alliance relationships, and loss of the capacity to project the governance values that define what kind of world order exists.

The Uncomfortable Logical Conclusion

If a nation that loses control of its oil supply faces economic collapse and potential state failure — recognized as justifying military response — and a nation that loses its strategic technology sector through systematic AI-enabled warfare faces the same terminal outcomes by the same logical chain, then the legal and strategic framework that justifies military response to oil supply threats should, by logical extension, apply to systematic AI-enabled economic destruction. It does not, because that framework was written before the capability existed. That gap is not a permanent feature of international law. It is an unresolved question that becomes more urgent with every passing year of AI capability development.

12.7 The International Law Framework and Its AI-Era Inadequacy

The Tallinn Manual — the authoritative academic treatment of international law applicable to cyber operations, produced by an international group of legal scholars at the invitation of the NATO Cooperative Cyber Defence Centre of Excellence — runs to two volumes and identifies 95 "black letter rules" governing cyber conflict. It represents the most serious existing attempt to apply international law to the cyber domain. It is inadequate for the AI era in several fundamental respects.

The Tallinn Manual's threshold analysis for what constitutes a "use of force" under Article 2(4) of the UN Charter focuses on effects — physical damage, death, destruction of infrastructure — rather than economic harm. The 1945 UN Charter drafting conference explicitly considered and rejected the inclusion of economic coercion as a "use of force," a decision that made sense in 1945 when economic warfare operated on timescales of years and physical destruction was the

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primary threat. That decision now creates a structural gap: systematic AI-enabled economic destruction that causes no physical damage, kills no one, destroys no building — but hollows out an entire nation's strategic industrial capacity — does not trigger the framework's recognition as armed aggression.

NATO has recognized that a serious cyberattack, or the cumulative impact of sustained malicious cyber activities, could invoke Article 5 — the collective defense clause requiring member nations to treat an attack on one as an attack on all. But the threshold for that recognition remains undefined, deliberately so, to preserve strategic ambiguity. As NATO's own analysis acknowledges, the question of when cyber operations cross the Article 5 threshold is "less clear" and the bar is assumed to be "very high." Adversaries who understand this ambiguity — and Chinese strategic doctrine in particular demonstrates sophisticated understanding of threshold management — exploit it systematically, operating permanently below whatever threshold might trigger collective response.

The Tallinn Manual 3.0 process, inaugurated in 2021 and spanning five years, is attempting to update the framework for new developments. It has not yet addressed the specific combination of AI speed, AI scale, AI attribution difficulty, and AI-enabled economic destruction magnitude that constitutes the threat this paper analyzes. It is, as of this writing, a framework in development for a threat already in deployment.

12.8 The Three Questions International Law Must Resolve

This paper does not advocate for military response to IP theft or economic warfare. That escalation ladder leads to catastrophic outcomes when nuclear-armed states are involved, and the attribution challenges of AI-enabled operations make proportional military response operationally problematic even where it might be legally justifiable. What this paper argues is more precise and ultimately more urgent: the international community must formally resolve three questions that AI-accelerated economic warfare makes impossible to defer.

12.8.1 The Recognition Question

At what cumulative economic damage threshold does AI-enabled economic warfare constitute an act of aggression under international law? The answer cannot be "never" — that creates unlimited license for economic destruction of any magnitude through AI means. It cannot be "always" — that conflates normal commercial competition with strategic warfare. It must be some threshold, defined in terms that are measurable, attributable, and proportionate to the severity of economic harm caused.

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Some European nations have begun to move toward explicit recognition. Norway and France have indicated that large-scale cyber operations against their national economy would qualify as uses of force. The Netherlands has raised the same prospect. The United States has not taken a clear position. The absence of a defined threshold is not strategic ambiguity — it is a unilateral disarmament of the response framework, benefiting only aggressors.

12.8.2 The Attribution Question

Who is responsible when AI operations are routed through multiple jurisdictions, executed by nominally private entities, and leave no forensic trail beyond compiled binaries distributed through third-country intermediaries? The laws of armed conflict require an identifiable state actor. AI-enabled economic warfare is specifically designed to defeat attribution. The Lazarus Group is nominally a criminal organization; only sustained intelligence community analysis confirmed North Korean state direction. Future AI-enabled operations, using domestically-hosted open source models with no US API logs, will be harder to attribute than Lazarus Group operations have been.

New attribution standards are necessary — potentially including presumptive attribution based on capability, pattern, and beneficiary analysis, similar to the intelligence community's "indicators and warnings" framework for state attribution of kinetic operations. An operation that benefits a state actor, is consistent with that actor's documented strategic objectives, employs capabilities consistent with that actor's known technical resources, and follows patterns consistent with that actor's historical operations — meets a preponderance standard for state attribution even without direct evidence of state direction.

12.8.3 The Response Question

If AI economic warfare is recognized as aggression and attributed to a state actor, what responses are proportionate and legal? This is the hardest question and the one the policy establishment most wants to avoid. The honest answer is that the current response toolkit — diplomatic protest, additional sanctions, civil litigation, export controls — is structurally inadequate against maximally-sanctioned state actors who have already built domestic AI capability. You cannot deter with sanctions an actor who faces maximum sanctions. You cannot litigate against a sovereign immune defendant. You cannot export-control a capability that already exists domestically.

This does not mean military response is appropriate. It means the response toolkit must be expanded beyond its current options to include capabilities that operate at the speed and scale of the threat. Offensive cyber operations — the US government's acknowledged but rarely

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discussed capability to impose costs on adversary digital infrastructure — are the most obvious candidate. The legal and policy framework for offensive cyber operations as a proportionate response to AI-enabled economic warfare needs to be developed and, to the extent possible, publicly articulated as a deterrence signal. Adversaries cannot be deterred by capabilities they do not know exist or policies they do not believe will be invoked.

12.9 The Deepest Strategic Reality

Beneath the legal analysis, beneath the policy proposals, beneath the threshold debates, lies a strategic reality that this paper's argument has been building toward from its first section: AI has fundamentally altered the relationship between technological capability and geopolitical power in a way that the international order has not yet recognized, incorporated, or begun to govern.

For seventy years, the post-WWII international order rested on a correlation between economic power and military power that made the order relatively stable. Large economies could sustain large militaries. Large militaries deterred aggression against large economies. Small, poor states were constrained in their ability to threaten the order's foundations. The nuclear arsenals of the great powers were the extreme expression of this correlation — concentrated destructive power in the hands of the wealthiest states, managed through mutual assured destruction.

AI breaks the correlation that stabilized the order. It allows states with minimal economic and military resources to project devastating economic force against the wealthiest nations. It compresses the timescale of economic destruction from decades to weeks. It defeats the attribution frameworks that make deterrence functional. And it exploits the gap between the speed of AI-enabled harm and the speed of legal, diplomatic, and institutional response — a gap that may be structural rather than merely transitional.

The question of whether economic warfare eventually becomes recognized as actual warfare — with the legal consequences, response rights, and alliance obligations that recognition entails — is not an IP law question. It is not even primarily a legal question. It is a question about what kind of international order humanity wants to maintain, whether the existing order's institutions are capable of adapting to AI-speed threats, and what happens to global stability if they are not.

This paper cannot answer those questions. They are beyond the scope of IP law reform, beyond the scope of export control policy, beyond the scope of any single discipline or any single whitepaper. What this paper can do — and what it has attempted to do throughout its twelve sections — is trace the logical chain from a specific legal gap in intellectual property doctrine all the way to its terminus in these existential strategic questions. That chain is real.

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The gaps in IP law are real. The governance failures are real. The state-sponsored threat is real. The geopolitical equalization that AI enables is real.

And the window for establishing frameworks — legal, technical, diplomatic, and strategic — that might govern this new reality before it governs us is closing. Not slowly. At machine speed.

13. Conclusions and Recommendations

13.1 Summary of Findings: A Stratified Assessment

This paper's analysis leads to conclusions that must be stated with the same stratification as the threat itself — organized by what is achievable in near, medium, and long terms, and what may never be fully addressed under any foreseeable governance framework.

Near Term (1-3 Years): What Can Actually Move

- Platform governance and AI provider acceptable use policies require no legislation and no international coordination. GitHub, major cloud providers, and AI API providers can implement provenance logging and reimplementation restrictions unilaterally. This is the only lever that operates at anything approaching machine speed, and it is available today.
- Trade secret law, robustly applied to AI-enabled reimplementation of disclosed proprietary code, provides the strongest existing legal claim against commercial misappropriation. Courts do not need new legislation to apply existing DTSA doctrine to AI-assisted misappropriation — they need cases, technical experts, and willingness to extend existing doctrine to new facts.
- Executive order authority under IEEPA exists today to extend export control frameworks to AI-assisted functional reimplementation of US-origin software in the state-sponsored threat context. This does not require Congressional action.
- Copyright registration of existing source code for $65 is available immediately and creates standing for statutory damages. This is low-hanging fruit available to every software company today.

Medium Term (3-7 Years): Legislative and Administrative Reform

- Fast-track Functional Architecture Registration — a new administrative mechanism providing affordable, rapid registration of software functional architecture — requires Congressional action but has a viable political pathway through the existing bipartisan consensus on China technology threat.

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- Expedited preliminary relief procedures for AI-enabled IP cases can be developed through court rulemaking and legislative amendment without requiring comprehensive IP reform.
- Provenance watermarking standards through NIST and IEEE can be developed and adopted within this timeframe, creating a forensic infrastructure that supports both commercial and national security enforcement.
- Explicit regulatory guidance under the Remote Access Security Act and existing EAR authority, specifying that AI-assisted functional reimplementation constitutes a controlled technology transfer, can extend the export control framework without new primary legislation.

Long Term (7-15 Years): Structural Reform

- Functional copyright doctrine — protecting the logical architecture of software systems against AI-assisted reproduction regardless of target language — requires either Supreme Court doctrinal evolution or comprehensive Congressional IP reform. This is a generational legislative undertaking.
- International coordination on AI provenance standards and mutual recognition of AI-era IP protections requires multilateral treaty negotiation on timescales measured in decades, not years.
- Recognition of AI-enabled economic warfare as a category of aggression under international law requires the development of new customary international law norms — a process that takes generations but begins with consistent state practice and the kind of explicit threshold articulation that nations like Norway and France have begun.

What May Never Be Fully Addressed

- State-sponsored AI-enabled economic warfare by maximally-sanctioned actors using domestically-developed open source AI capability against compiled binary targets with no forensic trail. Against this threat, IP law is irrelevant, export controls have limited reach, and the response toolkit requires capabilities — offensive cyber operations, credible deterrence signals, alliance coordination — that exist outside the legal framework this paper has analyzed.
- The fundamental power asymmetry that AI creates between small state offensive capability and large state defensive capacity. This asymmetry is a feature of the technology, not a bug correctable by governance. The international order must adapt to it or become progressively less capable of managing the conflicts it generates.

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13.2 Recommendations for Different Stakeholders

13.2.1 For Legislators

- Commission a comprehensive study of AI-enabled IP misappropriation, with technical expert input, as a foundation for legislative action.
- Introduce expedited preliminary relief procedures for AI-enabled IP cases.
- Explore a functional copyright regime through interagency coordination among the Copyright Office, USPTO, and NIST.
- Engage in bilateral and multilateral diplomatic efforts to establish international standards for AI provenance tracking and mutual recognition of AI-era IP protections.

13.2.2 For Courts

- Develop technical competency in AI systems and their capabilities, through court-appointed experts and dedicated technical staff.
- Apply trade secret law robustly in AI-enabled reimplementation cases, recognizing that the form of the reproduced information does not determine the scope of protection.
- Extend the doctrine of equivalents in patent cases to explicitly address AI-generated functional equivalents.
- Develop preliminary injunction standards appropriate for AI-velocity harms.

13.2.3 For Industry

- Adopt voluntary provenance tracking standards for AI systems capable of generating substantial code, as both ethical practice and litigation risk management.
- Establish clear internal policies against using AI systems to process or reimplement disclosed proprietary code from third parties.
- Engage constructively in legislative and regulatory processes to shape workable functional copyright standards.
- Invest in technical security practices to minimize unauthorized source code disclosure — including audit of source map exposure in production deployments.

13.2.4 For AI Developers

- Implement input filtering to detect and decline requests to process disclosed proprietary code for reimplementation purposes.
- Develop and publish provenance tracking capabilities that can be made available for legal proceedings.
- Engage with IP policy development proactively, rather than waiting for adverse legal outcomes to force reactive compliance.

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13.3 The Ultimate Conclusion

This paper began as an analysis of a specific legal gap: the inability of intellectual property law to reach AI-assisted functional reimplementation of proprietary software. It ends somewhere the authors of the Alice decision, the drafters of the DTSA, and the negotiators of TRIPS could not have anticipated when they built the frameworks this paper has analyzed. It ends at a question about the architecture of international order.

The chain of argument this paper has traced is worth stating in full, because each link is individually sound and the chain as a whole leads somewhere that demands acknowledgment:

Intellectual property law was built on three assumptions that AI has simultaneously destroyed. That copying requires significant human effort, providing natural deterrence. That translation between forms of expression requires skill and time sufficient to constitute independent creative work. And that similarity — the test for infringement — can be assessed by comparing surface-level expression that survives the translation process. When AI eliminates the effort, compresses the time to hours, and strips all surface expression through compilation to binary, all three assumptions fail at once. The doctrinal framework does not merely bend under this pressure. It collapses.

The collapse of IP doctrine is not merely a legal problem. It is a governance gap. And governance gaps — as the history of unrestricted warfare doctrine, state-sponsored economic espionage, and AI-enabled financial crime all demonstrate — are not neutral conditions. They are strategic opportunities for actors whose interests are served by the absence of enforceable rules. The gap in IP doctrine is being exploited today, at scale, by both commercial actors seeking competitive advantage and state-sponsored actors executing long-term campaigns of economic sector capture.

The governance gap is not merely an economic problem. When the mechanism of harm is AI-accelerated economic destruction — systematic, invisible, deniable, operating permanently below the threshold of recognized armed conflict — it becomes a security problem of the first order. A nation whose strategic technology sectors are systematically hollowed out by AI-enabled IP warfare loses not just revenue. It loses the industrial and intellectual infrastructure that sustains military capability, alliance credibility, and geopolitical power. The terminal outcome of unchecked AI economic warfare, prosecuted at sufficient scale and sustained over sufficient time, is indistinguishable from the terminal outcome of a sustained conventional military campaign against a nation's economic foundations.

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And the security problem is not merely a national problem. It is a problem of international order. AI has created the first technology in history that simultaneously democratizes devastating offensive economic capability — placing it within reach of any state actor and increasingly of non-state actors — while defeating the attribution, velocity, and jurisdictional prerequisites that all existing defensive frameworks were built to require. Every prior equalizing military technology, from gunpowder to nuclear weapons, was ultimately incorporable into deterrence frameworks because it operated in physical space where cause and effect were observable, attribution was achievable, and threshold recognition was possible. AI-enabled economic warfare operates in a domain where cause is invisible, attribution is defeated by architectural design, effect accumulates below any recognized threshold, and the timescale of harm is orders of magnitude faster than the timescale of any legal, diplomatic, or institutional response.

We are in a pre-framework moment. The technology of conflict has again outrun the institutional capacity to govern it — as it did in 1945 when nuclear weapons forced the construction of an entirely new international order, and as it did in 1999 when two PLA colonels wrote a doctrine for weaponizing that gap that is still being implemented today. The difference is that in 1945, the catastrophic visibility of nuclear weapons forced the international community to act. AI economic warfare leaves no mushroom cloud. Its damage accumulates quietly, company by company, sector by sector, below the threshold of recognition, below the threshold of response, below the threshold of alarm — until the damage is done and the barn door question is moot.

The window for establishing frameworks — legal, technical, diplomatic, and strategic — that might govern this new reality before it governs us is not infinite. It is not even comfortable. It is measured in years, not decades, because the behavior is normalizing faster than the governance is developing, and normalized behavior is the hardest kind to reverse.

The ultimate conclusion this paper reaches is this:

AI has made economic warfare as potentially existential as kinetic warfare — while making it simultaneously cheaper, faster, more deniable, and less governable than any form of conflict in human history. Intellectual property law is not equipped to address this. The international legal order is not equipped to recognize it. The governance frameworks that protect innovation, sustain competitive markets, and underwrite geopolitical stability were all built for a world in which this capability did not exist. That world is gone. The frameworks remain. The gap

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between them and the reality they were built to govern is where the most consequential conflicts of the next decade will be fought — quietly, invisibly, and at machine speed.

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Postscript: As This Paper Went to Press

April 1, 2026

This paper was completed on April 1, 2026. Within hours of its finalization, news broke of an event that renders every abstract argument in the preceding thirteen sections concrete, current, and urgent. It is documented here not as an addendum but as a timestamp — proof that the threat model this paper describes is not theoretical. It is operational. It is happening now.

INCIDENT: Axios npm Supply Chain Compromise

March 31 – April 1, 2026 | Attributed: UNC1069 (North Korea-nexus) | Google Threat Intelligence Group

Package: axios (npm) | Weekly downloads: ~100 million | Exposure window: 3 hours | Backdoor: WAVESHAPER.V2

On March 30, 2026 at 23:59 UTC, a threat actor compromised the npm account of the lead Axios maintainer — one of the most downloaded JavaScript libraries on earth, present in approximately 80% of cloud and code environments — and published two malicious package versions containing a hidden dependency. That dependency executed a cross-platform dropper automatically on every npm install, silently deploying the WAVESHAPER.V2 remote access trojan to Windows, macOS, and Linux systems simultaneously. The dropper then contacted a command-and-control server, delivered its payload, and deleted itself — replacing its own configuration file with a clean decoy so that any developer inspecting their installation after the fact would see nothing wrong.

The entire process from installation to full system compromise took approximately 15 seconds.

Google's Threat Intelligence Group attributed the attack to UNC1069, a financially motivated North Korea-nexus threat actor active since at least 2018, previously used by Pyongyang to target cryptocurrency and decentralized finance firms. The attribution was based on infrastructure overlaps with prior UNC1069 operations and the use of an updated version of the WAVESHAPER backdoor previously associated with the group.

The malicious versions were detected by automated security tooling and removed within roughly three hours. Three hours. Axios has approximately 100 million weekly downloads. Wiz estimated the malicious versions reached approximately 3% of the environments it scanned

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during the exposure window — a figure whose absolute scale, across the entire JavaScript ecosystem, represents a compromise event of extraordinary potential reach.

This was not an isolated incident. In the same week, a separate North Korea-linked group designated UNC6780 poisoned GitHub Actions and PyPI packages associated with the Trivy security scanner, the Checkmarx static analysis tool, and the LiteLLM AI library — deploying the SANDCLOCK credential stealer across an entirely different segment of the development infrastructure. Mandiant's CTO Charles Carmakal stated publicly that the secrets stolen across these simultaneous supply chain attacks "will enable more software supply chain attacks, SaaS environment compromises, ransomware and extortion events, and crypto heists over the next several days, weeks, and months."

What This Incident Proves

Section 10.2 of this paper described "supply chain infiltration through AI-generated compromised code" as one of five AI-enabled economic warfare weapon categories, and specifically cited the SolarWinds attack as a preview of what AI-automated versions at scale would look like. The Axios compromise is not a preview. It is the thing itself, executed yesterday, against infrastructure so foundational that 80% of cloud environments contain it. The 15-second compromise window. The self-deleting evidence. The simultaneous multi-vector campaign across npm, PyPI, and GitHub Actions. The North Korean state attribution. Every element of the threat model this paper constructed from first principles manifested in real time within 24 hours of the paper's completion.

Section 11.3 of this paper asked what happens when a small innovative company is driven out of business by AI-assisted economic warfare before its legal remedies can materialize. The Axios incident inverts that question. The victim here is not a company. The victim is the trust infrastructure of global software development itself — the assumption, foundational to the entire open source ecosystem, that a package downloaded from a reputable registry under a trusted maintainer's account contains what it says it contains.

That assumption was false for three hours on March 31, 2026. In those three hours, an unknown number of developer workstations, CI/CD pipelines, cloud build environments, and production systems across every industry that runs JavaScript — which is to say, essentially every industry — received a silent, self-concealing backdoor at the direction of the North Korean state. The downstream consequences, as Mandiant noted, will unfold over months.

Section 12 of this paper argued that we are in a pre-framework moment — that the technology of conflict has outrun the institutional capacity to govern it, and that the window for establishing

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governance frameworks is measured in years, not decades. The Axios incident compresses that timeline further. The question is not whether these attacks will continue. They will. The question is whether the frameworks — legal, technical, diplomatic, and strategic — will be established before the accumulated damage reaches a scale that forces recognition of what it actually is: warfare, conducted below every threshold that existing law and doctrine was built to recognize, against the foundations of the digital economy that every nation now depends upon for its security and prosperity.

The paper's ultimate conclusion stands. It is now simply more urgent than it was yesterday.

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References and Legal Citations

Editorial Note on Statistical Citations:

The $225B-$600B annual figure for Chinese IP theft originates from the Commission on the Theft of American Intellectual Property (2017 update). This figure has been cited extensively by the USTR and US government but has been subject to methodological critique — notably by the Niskanen Center and GAO — regarding the underlying data sources and attribution methodology. It is presented here as the most widely cited official estimate, not as a precisely

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validated figure. Readers engaged in policy work should consult the primary Commission report and its critiques before relying on specific dollar amounts. The cryptocurrency theft figures attributed to North Korea's Lazarus Group are drawn from Chainalysis, Elliptic, and TRM Labs blockchain analytics reports, cross-referenced against FBI attribution statements, and are considered reliable within the constraints of blockchain forensics methodology.

Primary Legal Authorities:

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